

Statistics 6910, Section 004, Fall 2013, Quiz 3 (25 Points)

Friday, December 6, 2013

Your Name: _____

Question 1: XML (18 Points)

The following example of an XML file has been found at http://www.flashkid.org/Tutorials/using_xml_in_flash_tutorial.html:

```
<Library>
  <Book id="1">
    <Title>Using XML with Flash</Title>
    <Authors>
      <Author>Salman Awan</Author>
      <Author>Christina Ray</Author>
    </Authors>
  </Book>
  <Book id="2">
    <Title>Book 2 Title</Title>
    <Authors>
      <Author>Single Author</Author>
    </Authors>
  </Book>
</Library>
```

1. (12 Points) Sketch an XML tree for this XML file. Use dashed lines for the terminal nodes and solid lines for the inner nodes.

2. (3 Points) Assume that this XML file is locally stored as `Quiz3_XML_Example.xml`. What will be the return value of the `xmlValue` function call below?

```
> setwd("C://JUE//Teaching//Stat6910_Fa2013_AdvancedR//Quizzes")
> library(XML)
> books = xmlParse("Quiz3_XML_Example.xml")
> root = xmlRoot(books)
> oneBook = root[[2]]
> xmlValue(oneBook[["Title"]])
```

3. (3 Points) Assume that the `xpathSApply` function call below produces the following output:

```
> xpathSApply(books, myXPath, xmlValue)

[1] "Salman Awan"    "Christina Ray" "Single Author"
```

What must have been the content of `myXPath` to obtain this result?

`myXPath =`

Question 2: SQL (7 Points)

1. (2 Points) What does SQL stand for?

2. (2 Points) What does RDBMS stand for?

3. (3 Points) Indicate two benefits of using a central database.